DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

ShopSphere – Online E-Commerce Database Management System

OBJECTIVES

1. Apply COUNT(), SUM(), AVG(), MIN(), and MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

1. AGGREGATE FUNCTIONS

1.1 COUNT()

Find the total number of orders:

```
SQL> SELECT COUNT(Order_ID) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           4
```

1.2 SUM()

Find the total sales amount:

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
       8495
```

1.3 AVG()

Find the average order amount:

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM Orders;

AVERAGE_ORDER_AMOUNT
--------------------
             2123.75
```

1.4 MIN()

Find the lowest order amount:

```
SQL> SELECT MIN(Total_Amount) AS Minimum_Order
  2  FROM Orders;

MINIMUM_ORDER
-------------
          899
```

1.5 MAX()

Find the highest order amount:

```
SQL> SELECT MAX(Total_Amount) AS Maximum_Order
  2  FROM Orders;

MAXIMUM_ORDER
-------------
         3598
```

2. GENERATE TOTAL SALES REPORTS

Generate a complete sales report using aggregate functions:

```
SQL> SELECT
  2      COUNT(Order_ID) AS Total_Orders,
  3      SUM(Total_Amount) AS Total_Sales,
  4      AVG(Total_Amount) AS Average_Order_Value,
  5      MIN(Total_Amount) AS Minimum_Order,
  6      MAX(Total_Amount) AS Maximum_Order
  7  FROM Orders;

TOTAL_ORDERS TOTAL_SALES AVERAGE_ORDER_VALUE MINIMUM_ORDER MAXIMUM_ORDER
------------ ----------- ------------------- ------------- -------------
           4        8495             2123.75           899          3598
```

2.1 Customer-wise Sales Report

Find how much each customer has purchased:

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      SUM(O.Total_Amount) AS Total_Purchase
  5  FROM Customer C
  6  JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.Customer_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        104 Divya Sri                                                    3598
        102 Priya Sharma                                                 2499
        101 Arun Kumar                                                   1499
        103 Rahul Raj                                                     899
```

3. FIND TOP CUSTOMERS BASED ON PURCHASE AMOUNT

Display customers according to their total purchase amount, from highest to lowest:

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      SUM(O.Total_Amount) AS Total_Purchase
  5  FROM Customer C
  6  JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID
  8  GROUP BY C.Customer_ID, C.Customer_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        104 Divya Sri                                                    3598
        102 Priya Sharma                                                 2499
        101 Arun Kumar                                                   1499
        103 Rahul Raj                                                     899
```

Purpose:

This query:

1. Connects Customer and Orders tables.
2. Groups records according to customers.
3. Calculates the total purchase amount.
4. Displays customers from highest to lowest purchase amount.

4. IDENTIFY BEST-SELLING PRODUCTS

Find the total quantity sold for each product:

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      SUM(OD.Quantity) AS Total_Quantity_Sold
  5  FROM Product P
  6  JOIN Order_Details OD
  7  ON P.Product_ID = OD.Product_ID
  8  GROUP BY P.Product_ID, P.Product_Name
  9  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       104 Mobile Phone
                   3

       103 Keyboard
                   1

       101 Laptop
                   1


PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       102 Mouse
                   1
```

Purpose:

This query:

1. Connects Product and Order_Details tables.
2. Groups records according to products.
3. Calculates the total quantity sold.
4. Displays products from highest to lowest quantity sold.

4.1 Product Sales Amount

Find the total sales amount generated by each product:

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      SUM(OD.Quantity) AS Total_Quantity_Sold,
  5      SUM(OD.Quantity * OD.Unit_Price) AS Total_Sales
  6  FROM Product P
  7  JOIN Order_Details OD
  8      ON P.Product_ID = OD.Product_ID
  9  GROUP BY
 10      P.Product_ID,
 11      P.Product_Name
 12  ORDER BY Total_Sales DESC;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       104 Mobile Phone
                   3        4947

       102 Mouse
                   1        2499

       101 Laptop
                   1        1499


PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
       103 Keyboard
                   1         899
```

5. CATEGORY-WISE SALES ANALYSIS

Find the total quantity sold and total sales generated by each product category:

```
SQL> SELECT
  2      C.Category_ID,
  3      C.Category_Name,
  4      SUM(OD.Quantity) AS Total_Quantity_Sold,
  5      SUM(OD.Quantity * OD.Unit_Price) AS Total_Sales
  6  FROM Category C
  7  JOIN Product P
  8      ON C.Category_ID = P.Category_ID
  9  JOIN Order_Details OD
 10      ON P.Product_ID = OD.Product_ID
 11  GROUP BY
 12      C.Category_ID,
 13      C.Category_Name
 14  ORDER BY Total_Sales DESC;

CATEGORY_ID
-----------
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES
------------------- -----------
          1
Electronics
                  6        9844
```

Purpose:

This query:

1. Connects Category, Product and Order_Details tables.
2. Groups records according to product categories.
3. Calculates the total quantity sold.
4. Calculates the total sales for each category.
5. Displays categories from highest to lowest sales.